

i. Background & Problem Statement:

LD (Lyme Disease) is caused by the spirochete bacteria *Borrelia burgdorferi* and is the most common tick-borne disease in the United States.¹ The disease is typically transmitted by a blacklegged tick (*Ixodes scapularis*) through a bite delivered to the host.² According to the Center for Disease Control (CDC), the number of individuals diagnosed with, and treated for, LD annually is estimated to approach approximately 476,000 cases.³ Interestingly, only 89,000 cases were officially reported to the CDC in 2023, meaning that the estimates include a probable ~400,000 cases being undiagnosed or misdiagnosed annually.³ Treatment typically consists of oral antibiotic therapy, however, intravenous antibiotics may be prescribed.⁴ Unfortunately, despite treatment, an estimated 10% of cases proceed to Post-Treatment LD Syndrome (PTLDS), which is a chronic condition featuring the hallmark symptoms of fatigue, sleep disturbances, pain, and cognitive impairments.⁵

Medical schools have benefitted from the discussions and research efforts in response to the LD epidemic, including expanded education, diagnostics, and treatment pathways.^{6,7} However, medical doctors are no longer the first point of contact for patients presenting with LD symptoms. Physical therapists (PT) and are providers who will frequently encounter patients presenting with LD symptoms before, or in lieu of, a medical doctor. Therefore, it is of critical importance that they be able differentially diagnose LD from other common conditions that can imitate the infection. For example, a swollen, stiff, and painful knee joint could be osteoarthritis or a joint sprain, but it also could be the manifestations of an untreated LD infection.⁸ As direct access providers of choice for musculoskeletal issues, it becomes paramount that a PT recognize these sometimes vague and transient symptoms as potential LD so a referral can be made for medical examination and treatment.

While medical schools have made progress, it is unclear if the PT profession has made the necessary curricular revisions to address their increased role in differentially diagnosing LD. Since we know LD is frequently under or misdiagnosed, and we know that PT's are first line providers, understanding the efforts being undertaken in the academic programs licensing these practitioners regarding LD and diagnosis is key. Additionally, there are no studies that currently examine a practicing PT's ability to differentially diagnose LD, and only one study specifically published on LD diagnosis and PT.⁹ Therefore, the purpose of this study is to survey educators about the curricular trends surrounding LD in PT programs, as well as survey clinicians to assess their ability to differentially diagnose LD. By identifying these factors, we can better advocate for the need for further educational interventions and better prepare future PT's for clinic practice.

ii. Specific Aims

There are two Aims of this study. Aim 1 is to explore curricular trends related to LD in accredited PT programs. Aim 2 is to assess differential diagnostic capabilities as it relates to LD among practicing PT's.

iii. Methodical Approach and Timeline

For Aim 1, a survey will be designed and content validated specific to the curricular trends in PT programs related to LD. The inclusion criteria for participating in the survey is that: 1) they are a faculty member or instructor in an accredited/candidate PT program, and 2) have knowledge of the curricular trends involving LD in their program. Recruitment will occur in three ways: posting to the Education Special Interest Group of the Academy of Education discussion board, direct email to the Chair/Program Director of all PT programs, and in-person data collection at the Combined Sections Meeting (CSM) in February of 2026. The study will continue until 100 participants are reached (the limit of requested funds and a sample of ~33% of all PT programs) before the survey will be closed. Only one participant from each program can complete the survey (a question on the survey will ask which program they are representing and the institution will be cross-referenced to prevent duplicates). In addition to collecting a demographic profile on each participant, the main themes to be explored are the amount of time in the program dedicated to LD education, the modes in which the education occurs, and degree of focus on differential diagnosis. The survey will be created by the research team, then content validated by at least two individuals unrelated to the study but who have expertise in the content area and survey design. The survey will be piloted prior to being deployed. Data will be analyzed using descriptive statistics and crosstabulation for the variable of geographic location. It is hypothesized that geographical location may influence the amount of LD education in a program due to the disease's preponderance to the particular locales. A \$10 gift card will be digitally sent to participants within 4 weeks of receiving a completed survey.

Aim 2 will be a combination of a survey and questionnaire. Inclusion criteria for the second Aim is that they are a licensed physical therapist practicing in an outpatient setting. They are excluded from participating in the study if they are faculty members in a PT program (to avoid possible identification of the study's purpose and redundancy among the samples). Recruitment will occur in the following ways: posting on social media, posting on the American Physical Therapy Association discussion boards, and in-person data collection at CSM in February of 2026. The study will continue until 100 participants are reached (the limit of requested funds) and the survey will be closed. The survey will collect demographic data and define the characteristics of the participating physical therapists. The questionnaire portion will be a series of case scenarios presenting a simulated patient encounter. The scenarios will require the participant to determine the differential diagnosis by listing (in order of likelihood) the top 3 diagnoses they believe to be presented by the scenario. They will then be asked to choose their primary diagnosis and why. The survey and questionnaire will not mention LD as this would disclose the purpose of the study and influence results. The survey and questionnaire for Aim 2 will be content validated by experts in test writing and piloted before deployment. Data analysis will explore the demographics and answers to the questionnaire using descriptive statistics, and compare the responses for accuracy based on geographical location and setting. It is hypothesized that there will be a difference in the diagnosis of LD based on the region of the participant completing the questionnaire. A \$20 gift card will be digitally sent to participants within 2 weeks of receiving a completed survey. The increased reimbursement from Aim 1 is due to the estimated increase in time required to complete the study.

The timeline for completing the project is as follows:

2025				2026							
September	October	November	December	January	February	March	April	May	June	July	August
Design surveys and questionnaire											
		Content validate & Pilot									
			Submit IRB (exempt)								
				Data collection							
					In person data collection						
									Data analysis		
										Submission for poster/platform at CSM 2027	
											Manuscript preparation
				Data monitoring and payouts							

iv. Potential Study Impact

Each aim individually addresses important concerns affecting PT practice. The results of Aim 1 can be used as a baseline to draw comparisons across the hundreds of different PT programs nationwide. Programs can benchmark themselves against the results of the study in order to address shortfalls in their curriculum. It may also serve as a repository of potential teaching methodologies for educators to utilize in their own programs.

The results of Aim 2 will address the question of a PT's ability to differentially diagnose LD from other common imitators, shedding light on the degree of underdiagnosis present among clinicians. This knowledge will bring to light gaps in clinical practice, allowing for discussion and advocacy efforts to have a basis in evidence.

Taking the two Aims together, a strong case for the need to reform PT education and practice can be made. If the studies correctly identify that there is a paucity of education on LD differential diagnosis, and that we have evidence of misdiagnosis clinically, then the next logical step is to address the issue through educational interventions directed at both PT students and practicing clinicians.

Furthermore, future studies could expand the scope of practice beyond physical therapists and examine the curricular trends and diagnostic capabilities in other fields, including but not limited to physician assistants and nurse practitioners.

v. Roles and responsibilities of PI and Co-Investigators

The PI in this study is responsible for the coordination of all elements of the study. This includes the IRB, designing of the surveys and questionnaire, recruitment, data collection and analysis, and scientific writing and dissemination.

Co-I (GS) is a physician assistant faculty member within SHP and will serve as the content expert and lead the efforts in designing the surveys and questionnaire. He will play a role in interpreting the results and writing of the manuscript. Co-I (CV) has experience in survey design and recruitment within PT education and will assist in designing of the surveys and data

collection (both in-person and remotely). Both will assist in the dissemination phase of the study.

The graduate assistant will assist in data collection, data management (assuring completed surveys are submitted, keeping a list of unique identifiers to prevent redundant submissions, and issuing payment to participants), as well as participate in the in-person data collection at CSM.

vi. Plans for Future Grant Submissions and Dissemination

Each of the aims are individual scholarly products consisting of a platform or poster presentation as well as a manuscript (a total of 4 scholarly products). The results of Aim 1 would be submitted for presentation at the Educator Leadership Conference (ELC), as well being submitted for publication in the Journal of Physical Therapy Education (JOPTE). The second Aim would be submitted for presentation at the Combined Sections Meeting (CSM) of the American Physical Therapy Association (APTA) and submitted for publication in the Journal of Physical Therapy (PTJ).

The combined results of the study are expected to indicate that there is both a lack of time spent addressing LD diagnosis in physical therapy education as well as a failure of practicing physical therapists to correctly identify LD when presented with case scenarios. The results of the study can be used to support the need for an interventional study aimed at physical therapist education and clinical practice. The international Lyme and Associated Diseases Educational Foundation (ILADEF) has an annual grant specifically designed to fund education related to LD (<https://iladef.org/grants-available/>). This includes the development and distribution of educational materials on tick borne diseases and other related dissemination costs.

vii. References

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